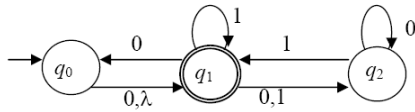


Midterm 1 – Regular Languages

Problem 1. (4 points) Consider the following NFA over the alphabet $\Sigma = \{0,1\}$:



a) Write the transition function for the NFA.

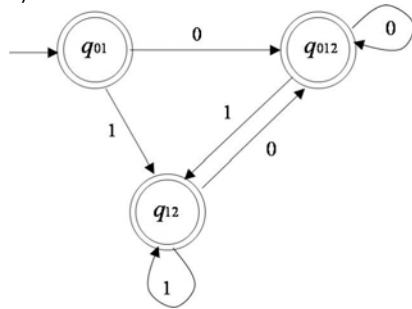
b) Convert the NFA into an equivalent DFA.

Solution 1:

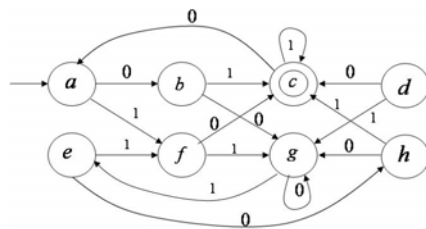
a)

δ	0	1	λ
q_0	$\{q_1\}$	-	$\{q_1\}$
q_1	$\{q_0, q_2\}$	$\{q_1, q_2\}$	-
q_2	$\{q_2\}$	$\{q_1\}$	-

b)

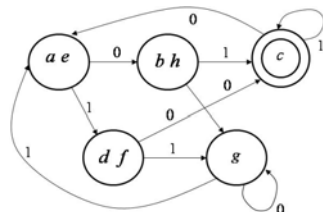


Problem 2. (4 points) Let M be the following DFA.



Minimize M by set partitioning.

Solution 2:



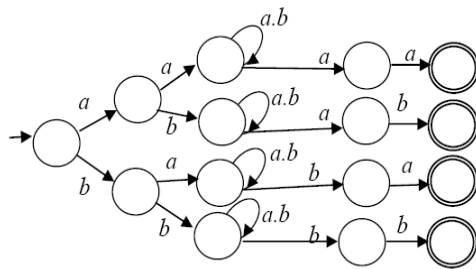
Problem 3 (6 points)

- a) Is the language described as follows: $L = \{ \text{The set of all strings over the alphabet } \{0\} \text{ of the form } 0^n \text{ where } n \text{ is not prime.} \}$ regular or not? If it is regular, construct an automaton, regular expression or a grammar. If not regular, use pumping lemma for regular languages.
- b) Show that the language $L = \{vwv : v, w \in \{a,b\}^*, |v| = 2\}$ is regular.
- c) Show that the language $L = \{w : n_a(w) = n_b(w)\}$ is not regular.

Solution 3:

a) $L = \{ \text{The set of all strings over the alphabet } \{0\} \text{ of the form } 0^n \text{ where } n \text{ is not prime.} \}$ **NOT REGULAR**. To prove this language is not regular, we instead examine the complement because the set of regular languages is closed under complement. We have shown (by Pumping lemma) in the class that the complement of the language (0^n where n is prime) is not regular. That means that language cannot be regular.

b) $L = \{vwv : v, w \in \{a,b\}^*, |v| = 2\}$ is **REGULAR**. Here is an automaton for L :



d) $L = \{w : n_a(w) = n_b(w)\}$ is **NOT REGULAR**. Justification:

The Pumping Lemma states that for any regular language there is a value m so that if you pick any string of the language with a length greater than m , then we can always find a non-empty portion y of the string that appears in the string within the first m characters that can be “pumped”, i.e. repeated as many times as we like, forming strings that are always in the language. The key here is to find a counterexample, a string in the language of length $\geq m$, for which it will be easy to show that there is no way to pump it.

Proof: Assume L is regular. Therefore the Pumping Lemma holds for L . Consider the string $a^m b^m$. This string is clearly longer than m and clearly belongs to the language because the number of a 's = the number of b 's. By the Pumping Lemma, this string can be represented as xyz with $|xy| \leq m$ and y non-empty, so that $xy^i z$ is in L for all values of i . But since the length of xy is $\leq m$ and there are m a 's at the start of the string, xy (and also y) must consist of only a 's. So $y = a^k$ with $k > 0$. String $xy^2 z$ is $a^{m+k} b^m$, because we have added another y (or a^k) to the string. But the number of a 's here does not equal the number of b 's, so this string is NOT in the language. This contradicts the Pumping Lemma. Therefore, since the Pumping Lemma holds for all regular languages, the language $L = \{w : n_a(w) = n_b(w)\}$ is not regular.

References

Linz Peter, An Introduction to Formal Languages and Automata, Jones & Bartlett, 2006

Salling Lennart: Formella språk, automater och beräkningar, 2001

Sipser Michael, Introduction to the Theory of Computation, PWS 1997